

Pandas Data Analysis Project

Project Objective

The objective of this project is to perform data analysis using Python and Pandas library. The project demonstrates loading CSV data, cleaning data, filtering records, grouping data, and generating insights.

Technologies Used

- Python
 - Pandas
 - Matplotlib
 - VS Code
-

Dataset Used

Supermarket Sales Dataset (CSV format)

Features Implemented

- Data Loading using Pandas
 - Data Cleaning
 - Missing Value Checking
 - Filtering Data
 - Grouping and Aggregation
 - Data Visualization using Bar Chart
 - Insight Generation
-

Code Snippet

```
import pandas as pd
import matplotlib.pyplot as plt

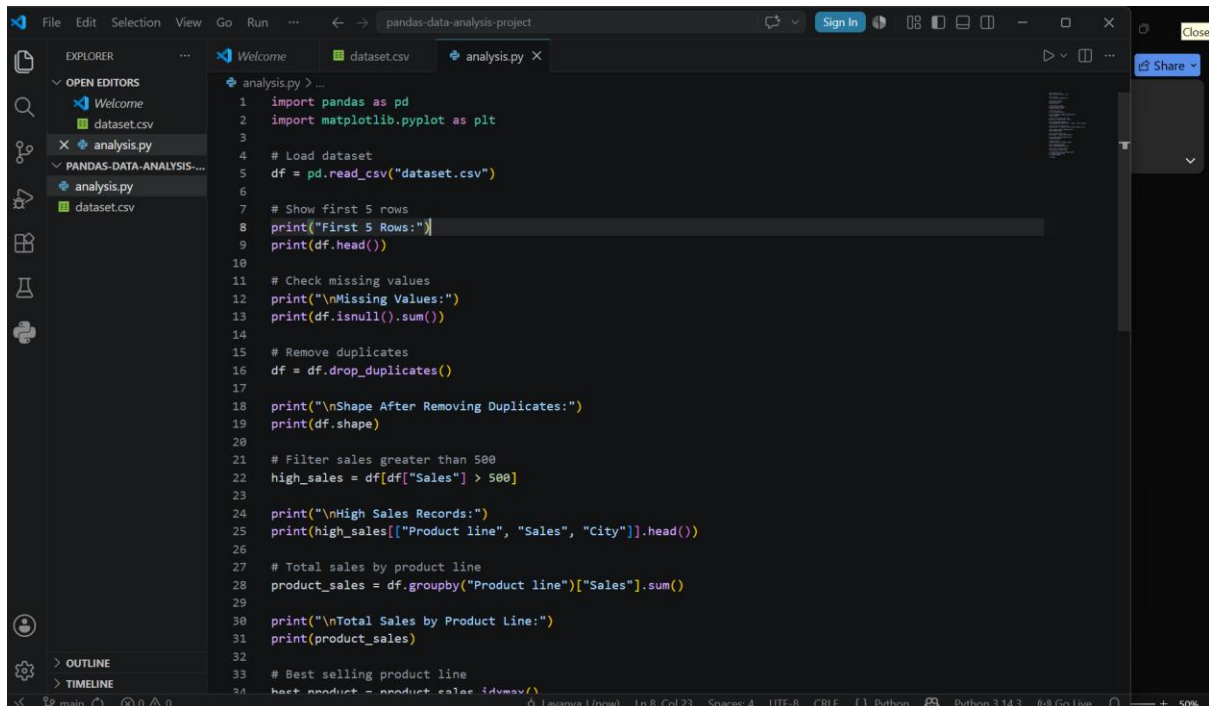
df = pd.read_csv("dataset.csv")
```

```
product_sales = df.groupby("Product line")["Sales"].sum()
```

```
product_sales.plot(kind="bar")
```

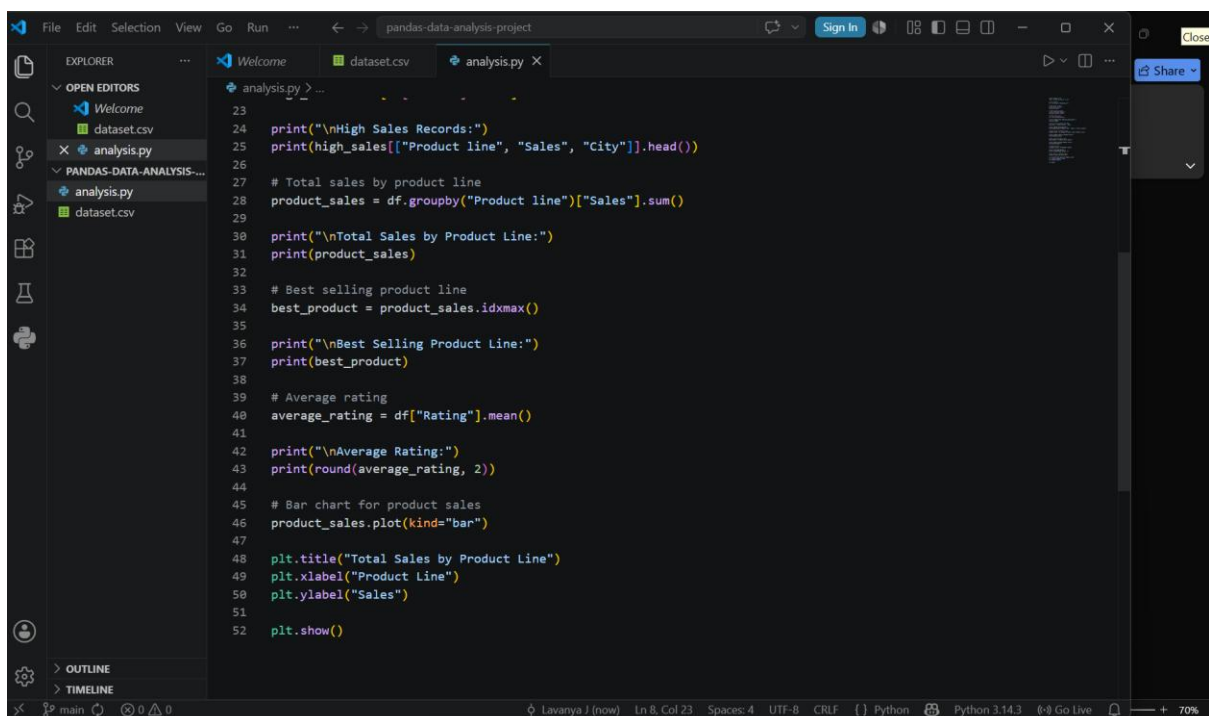
```
plt.show()
```

Code Screenshots



This screenshot shows a code editor with a file explorer on the left and a code editor on the right. The file explorer shows a project named 'pandas-data-analysis-project' with files 'dataset.csv' and 'analysis.py'. The code editor shows the following Python code:

```
1 import pandas as pd
2 import matplotlib.pyplot as plt
3
4 # Load dataset
5 df = pd.read_csv("dataset.csv")
6
7 # Show first 5 rows
8 print("First 5 Rows:")
9 print(df.head())
10
11 # Check missing values
12 print("\nMissing Values:")
13 print(df.isnull().sum())
14
15 # Remove duplicates
16 df = df.drop_duplicates()
17
18 print("\nShape After Removing Duplicates:")
19 print(df.shape)
20
21 # Filter sales greater than 500
22 high_sales = df[df["Sales"] > 500]
23
24 print("\nHigh Sales Records:")
25 print(high_sales[["Product line", "Sales", "City"]].head())
26
27 # Total sales by product line
28 product_sales = df.groupby("Product line")["Sales"].sum()
29
30 print("\nTotal Sales by Product Line:")
31 print(product_sales)
32
33 # Best selling product line
34 best_product = product_sales.idxmax()
```



This screenshot shows the same code editor with the second part of the Python script. The code continues from the previous screenshot:

```
23
24 print("\nHigh Sales Records:")
25 print(high_sales[["Product line", "Sales", "City"]].head())
26
27 # Total sales by product line
28 product_sales = df.groupby("Product line")["Sales"].sum()
29
30 print("\nTotal Sales by Product Line:")
31 print(product_sales)
32
33 # Best selling product line
34 best_product = product_sales.idxmax()
35
36 print("\nBest Selling Product Line:")
37 print(best_product)
38
39 # Average rating
40 average_rating = df["Rating"].mean()
41
42 print("\nAverage Rating:")
43 print(round(average_rating, 2))
44
45 # Bar chart for product sales
46 product_sales.plot(kind="bar")
47
48 plt.title("Total Sales by Product Line")
49 plt.xlabel("Product Line")
50 plt.ylabel("Sales")
51
52 plt.show()
```

Output Screenshots

VS Code terminal window showing the output of a Python script. The script is running in a PowerShell session. The output displays the first 5 rows of a dataset and the missing values for each column.

```
PS C:\Users\Lavanya J\OneDrive\Desktop\pandas-data-analysis-project> python -u "c:\Users\Lavanya J\OneDrive\Desktop\pandas-data-analysis-project\analysis.py"
First 5 Rows:
  Invoice ID Branch      City Customer type Gender ... Payment  cogs  gross margin percentage  gross income  Rating
0  750-67-8428  Alex      Yangon      Member Female ...   Ewallet  522.83           4.761985           26.1415     9.1
1  226-31-3881  Giza      Naypyitaw      Normal Female ...    Cash    76.40           4.761985           3.8280     9.6
2  631-41-3108  Alex      Yangon      Normal Female ... Credit card  324.31           4.761985           16.2155     7.4
3  123-19-1176  Alex      Yangon      Member Female ...   Ewallet  465.76           4.761985           23.2880     8.4
4  373-73-7910  Alex      Yangon      Member Female ...   Ewallet  604.17           4.761985           30.2085     5.3

[5 rows x 17 columns]

Missing Values:
Invoice ID      0
Branch          0
City            0
Customer type   0
Gender          0
Product line    0
Unit price      0
Quantity        0
Tax 5%          0
Sales           0
Date            0
Time            0
Payment         0
cogs            0
gross margin percentage  0
gross income    0
Rating          0
dtype: int64

Shape After Removing Duplicates:
(1000, 17)

High Sales Records:
  Product line  Sales      City
0  Health and beauty  548.9715  Yangon
```

VS Code terminal window showing the output of a Python script. The script is running in a PowerShell session. The output displays high sales records and total sales by product line.

```
Date          0
Time          0
Payment       0
cogs          0
gross margin percentage  0
gross income  0
Rating        0
dtype: int64

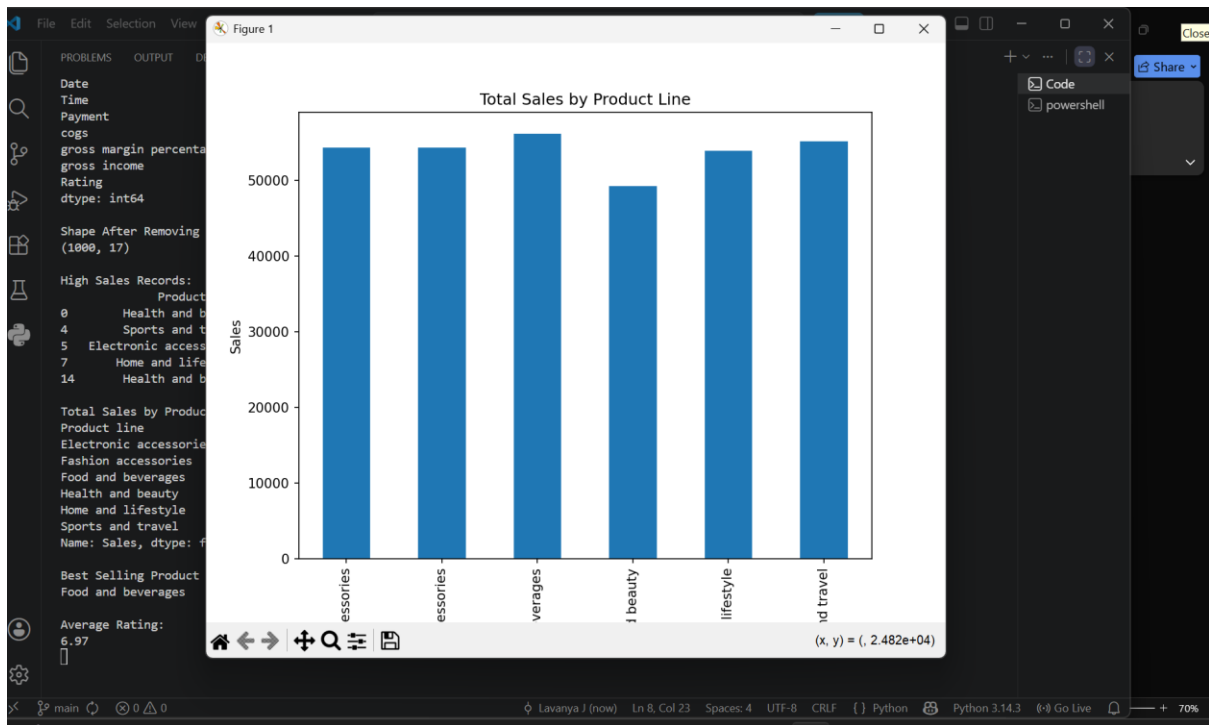
Shape After Removing Duplicates:
(1000, 17)

High Sales Records:
  Product line  Sales      City
0  Health and beauty  548.9715  Yangon
4  Sports and travel  634.3785  Yangon
5  Electronic accessories  627.6165  Naypyitaw
7  Home and lifestyle  772.3880  Naypyitaw
14 Health and beauty  749.4980  Yangon

Total Sales by Product Line:
Product line
Electronic accessories  54337.5315
Fashion accessories    54305.8950
Food and beverages     56144.8440
Health and beauty      49193.7390
Home and lifestyle     53861.9130
Sports and travel      55122.8265
Name: Sales, dtype: float64

Best Selling Product Line:
Food and beverages

Average Rating:
6.97
```



Insights

1. Product categories generated different levels of sales.
 2. High-value sales transactions were identified.
 3. Average customer ratings were good.
 4. Dataset contained no missing values.
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GitHub Repository Link

<https://github.com/jlavanya132008-byte/pandas-data-analysis-pro>

Hosted Link

<https://pandas-data-analysis-project-1.onrender.com>

Conclusion

This project helped in understanding practical data analysis using Pandas and visualization using Matplotlib.